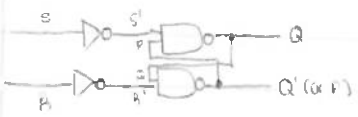
NAND implementation

Active-high SR latch



S	R	Q	Q'	
1	0	1	0	
0	1	0	1	
0	0	1	0	after (SR=10)
0	0	0	1	after (SR=01)
1	1			(not allowed)
1	1			

$Q = (Q'S')' = (PS')' = P' + S = Q + S$

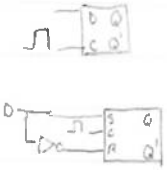
$Q' = (QR)' = Q' + R$

Control input

- Controls when the state change can happen
- enable
- NAND latch

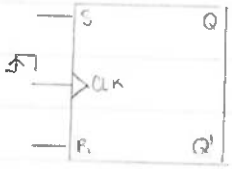
} SR Latch

D	C	Q	Q'
0	1	0	1
1	1	1	0
X	0	Q_n	Q_n'



SR Flip-flop characteristic table

C	S	R	Q_{n+1}	Q'_{n+1}	function
↑	0	0	Q_n	Q'_n	no change
↑	0	1	0	1	reset
↑	1	0	1	0	set
↑	1	1	0	0	forbidden
0	X	X	Q_n	Q'_n	inhibited



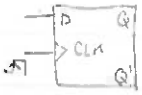
Q_n : state before positive edge

Q_{n+1} : state after positive edge

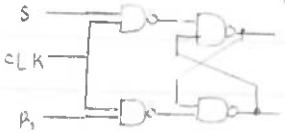
D Flip-flop

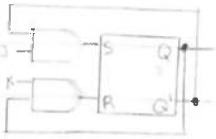
D	CLK	Q_{n+1}
0	↑	0
1	↑	1
X	0	Q_n

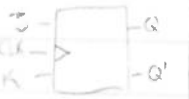
↑ characteristic table



JK Flip-flop







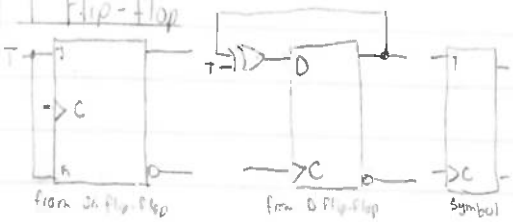
J	K	Q_{n-1}	Function
0	0	Q_n	no change
0	1	0	reset
1	0	1	set
1	1	Q'_n	complement (also toggle)

If $Q_n = 1$, $Q_{n+1} = 0$

If $Q_n = 0$, $Q_{n+1} = 1$

→ characteristic table

T Flip-flop



CLK	T	Q_{n+1}
0	X	Q_n (memory)
↓	0	Q_n (memory)
↓	1	Q_n' (toggle)